

## ENTHUSIAST & LEADER COURSE

### TARGET : PRE-MEDICAL 2020

Test Type : Major

Test # 01

Test Pattern : NEET-UG

TEST DATE : 17 - 04 - 2020

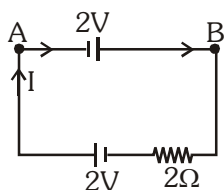
### ANSWER KEY

|      |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |     |
|------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| Que. | 1   | 2   | 3   | 4   | 5   | 6   | 7   | 8   | 9   | 10  | 11  | 12  | 13  | 14  | 15  | 16  | 17  | 18  | 19  | 20  |
| Ans. | 1   | 2   | 3   | 2   | 2   | 4   | 3   | 1   | 1   | 4   | 3   | 4   | 1   | 1   | 3   | 1   | 2   | 2   | 4   | 2   |
| Que. | 21  | 22  | 23  | 24  | 25  | 26  | 27  | 28  | 29  | 30  | 31  | 32  | 33  | 34  | 35  | 36  | 37  | 38  | 39  | 40  |
| Ans. | 4   | 1   | 1   | 1   | 2   | 3   | 1   | 1   | 2   | 2   | 4   | 1   | 1   | 2   | 2   | 4   | 1   | 4   | 1   | 1   |
| Que. | 41  | 42  | 43  | 44  | 45  | 46  | 47  | 48  | 49  | 50  | 51  | 52  | 53  | 54  | 55  | 56  | 57  | 58  | 59  | 60  |
| Ans. | 2   | 1   | 1   | 4   | 2   | 2   | 3   | 1   | 2   | 1   | 4   | 1   | 4   | 1   | 1   | 4   | 1   | 2   | 2   | 1   |
| Que. | 61  | 62  | 63  | 64  | 65  | 66  | 67  | 68  | 69  | 70  | 71  | 72  | 73  | 74  | 75  | 76  | 77  | 78  | 79  | 80  |
| Ans. | 2   | 4   | 1   | 3   | 2   | 3   | 1   | 2   | 2   | 2   | 1   | 1   | 4   | 2   | 3   | 2   | 3   | 2   | 2   | 2   |
| Que. | 81  | 82  | 83  | 84  | 85  | 86  | 87  | 88  | 89  | 90  | 91  | 92  | 93  | 94  | 95  | 96  | 97  | 98  | 99  | 100 |
| Ans. | 2   | 2   | 2   | 2   | 1   | 3   | 2   | 3   | 4   | 1   | 2   | 1   | 2   | 3   | 2   | 2   | 1   | 2   | 2   | 2   |
| Que. | 101 | 102 | 103 | 104 | 105 | 106 | 107 | 108 | 109 | 110 | 111 | 112 | 113 | 114 | 115 | 116 | 117 | 118 | 119 | 120 |
| Ans. | 2   | 4   | 3   | 1   | 3   | 4   | 1   | 3   | 2   | 2   | 1   | 3   | 3   | 3   | 2   | 2   | 4   | 3   | 4   | 4   |
| Que. | 121 | 122 | 123 | 124 | 125 | 126 | 127 | 128 | 129 | 130 | 131 | 132 | 133 | 134 | 135 | 136 | 137 | 138 | 139 | 140 |
| Ans. | 3   | 1   | 3   | 4   | 3   | 1   | 1   | 3   | 3   | 2   | 4   | 2   | 2   | 3   | 1   | 4   | 2   | 2   | 1   | 2   |
| Que. | 141 | 142 | 143 | 144 | 145 | 146 | 147 | 148 | 149 | 150 | 151 | 152 | 153 | 154 | 155 | 156 | 157 | 158 | 159 | 160 |
| Ans. | 1   | 1   | 2   | 1   | 4   | 3   | 3   | 3   | 3   | 2   | 1   | 3   | 3   | 3   | 2   | 2   | 2   | 2   | 3   | 3   |
| Que. | 161 | 162 | 163 | 164 | 165 | 166 | 167 | 168 | 169 | 170 | 171 | 172 | 173 | 174 | 175 | 176 | 177 | 178 | 179 | 180 |
| Ans. | 4   | 4   | 2   | 4   | 4   | 1   | 2   | 4   | 4   | 1   | 1   | 1   | 1   | 4   | 2   | 2   | 2   | 2   | 1   | 2   |

### HINT - SHEET

1. Component of  $\vec{A}$  on  $\vec{B} = \frac{\vec{A} \cdot \vec{B}}{|\vec{B}|} = \frac{2+3}{\sqrt{2}} = \frac{5}{\sqrt{2}}$

2. Circuit can be drawn as



Total emf = 2 + 2 = 4V

so  $I = \frac{4}{2} = 2A$

4. Amplitude of electric field and magnetic field are related by the relation

$$\frac{E_0}{B_0} = c$$

Average energy density of the magnetic field is

$$u_B = \frac{1}{4} \frac{B_0^2}{\mu_0}$$

$$= \frac{1}{4} \frac{E_0^2}{\mu_0 c^2} \quad \left( \because B_0 = \frac{E_0}{c} \right)$$

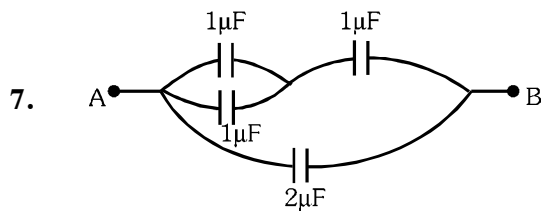
$$= \frac{1}{4} \epsilon_0 E_0^2 \quad \left( \because c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \right)$$

$$= \frac{1}{4} \times 8.854 \times 10^{-12} \times (2)^2$$

$$\approx 8.86 \times 10^{-12} \text{ J m}^{-3}$$

5.  $\beta = \frac{\lambda D}{d} \Rightarrow \beta \propto \frac{1}{d}$

6.  $P = 50 \frac{\text{N}}{\text{m}^2} = 50 \times \frac{10^5}{(10^2)^2} \frac{\text{dyne}}{\text{cm}^2} = 500 \frac{\text{dyne}}{\text{cm}^2}$



Redraw the ckt.  $C_{AB} = \frac{8}{3} \mu\text{F}$

8. Force on wire Q due to wire P is

$$F_P = 10^{-7} \times \frac{2 \times 30 \times 10}{0.1} \times 0.1 = 6 \times 10^{-5} \text{ N}$$

(Towards left)

Force on wire Q due to wire R is

$$F_R = 10^{-7} \times \frac{2 \times 20 \times 10}{0.02} \times 0.1 = 20 \times 10^{-5} \text{ N}$$

(Towards right)

Hence  $F_{\text{net}} = F_R - F_P = 14 \times 10^{-5} \text{ N} = 1.4 \times 10^{-4} \text{ N}$

(Towards right)

9. Revolutions per minute (N) =  $\frac{3000}{\pi} \text{ rpm}$

so, frequency,  $n = \frac{N}{60} = \frac{3000}{60 \times \pi} = \frac{50}{\pi} \text{ Hz}$

and  $\omega = 2\pi n = 2\pi \times \frac{50}{\pi} = 100 \text{ rad/sec}$

$I = 400 \text{ kg m}^2$

so rotational kinetic energy (K) is

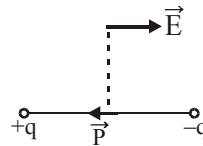
$$K = \frac{1}{2} I \omega^2 = \frac{1}{2} \times 400 \times (100)^2 = 2 \times 10^6 \text{ Joule}$$

10.  $\frac{I_{\text{max}}}{I_{\text{min}}} = \left( \frac{a_1 + a_2}{a_1 - a_2} \right)^2 = \frac{36}{1}$

$$\Rightarrow \frac{a_1 + a_2}{a_1 - a_2} = \frac{6}{1}$$

$$\Rightarrow \frac{a_1}{a_2} = \frac{7}{5}$$

11.



12.  $C = \frac{\epsilon_0 A}{d}$ ;  $C_1 = \frac{\epsilon_0 A}{2d} = \frac{C}{2}$

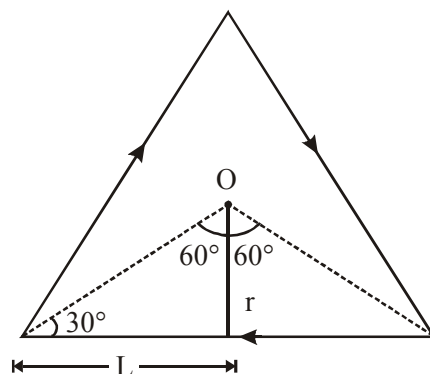
$$C_2 = \frac{5 \epsilon_0 A}{2d} = \frac{5}{2} C$$

$$C_{\text{eq}} = C_1 + C_2 = 3C$$

$$\frac{\Delta C}{C} \% = \frac{2C}{C} \times 100 = 200\%$$

13. Magnetic field at O is

$$B = 3 \left[ \frac{\mu_0 i}{4\pi r} (\sin 60^\circ + \sin 60^\circ) \right]$$



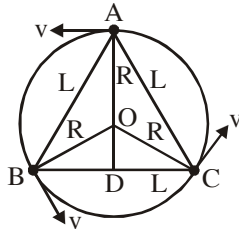
$$\tan 30^\circ = \frac{r}{L} \Rightarrow r = \frac{L}{\sqrt{3}}$$

$$= 3 \left( \frac{\mu_0}{4\pi} \right) \left( \frac{i}{L} \right) \left( \frac{\sqrt{3}}{2} \right) (\sqrt{3}) \times 2$$

$$= \frac{9\mu_0 i}{4\pi L}$$

14.  $\vec{v} = \vec{\omega} \times \vec{r} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -5 & 3 & 5 \\ 3 & 4 & 6 \end{vmatrix} = -2\hat{i} + 45\hat{j} - 29\hat{k}$

15. Consider the circle with centre at O, and having radius R. In it consider three bodies of masses M each moving with velocity v under the action of their gravitational attraction.



$$R = \frac{2}{3} AD = \frac{2}{3} \times L \sin 60^\circ = \frac{2L}{3} \times \frac{\sqrt{3}}{2}$$

$$= \frac{L}{\sqrt{3}} \quad \text{or} \quad L = \sqrt{3}R$$

Centripetal force on any one mass M is

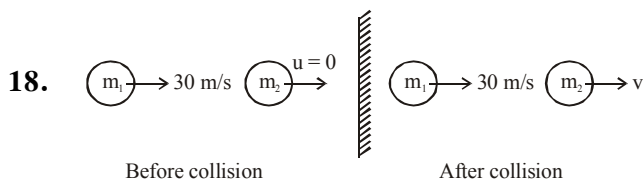
$$= \frac{2GM^2}{L^2} \cos\left(\frac{60^\circ}{2}\right)$$

$$\therefore \frac{Mv^2}{R} = \frac{2GM^2}{L^2} \times \frac{\sqrt{3}}{2}$$

$$\text{or } \frac{GM^2}{(\sqrt{3}R)^2} \sqrt{3} = \frac{Mv^2}{R}$$

$$\therefore v = \sqrt{\frac{GM}{\sqrt{3}R}}$$

16.  $E = -\frac{dV}{dx} = -(10x + 10) = -(10 + 10)$   
 $= -20 \text{ Vm}^{-1}$



Since  $m_1 \gg m_2$ , therefore heavy body ( $m_1$ ) does not change its velocity after collision. By the definition of coefficient of restitution

$$e = \frac{v_2 - v_1}{u_1 - u_2} \Rightarrow 1 = \frac{v - 30}{30 - 0}$$

$$\Rightarrow v = 60 \text{ m/s}$$

19.  $\frac{\theta_1 - \theta_2}{t} = -K \left( \frac{\theta_1 + \theta_2}{2} - \theta_0 \right)$

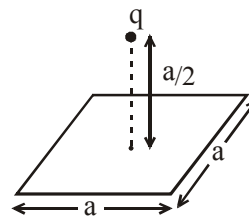
$$t = 2 \text{ min} = 120 \text{ s}$$

$$\frac{80 - 70}{120} = -K \left( \frac{80 + 70}{2} - 30 \right) \dots (i)$$

$$\frac{60 - 50}{t} = -K \left( \frac{60 + 50}{2} - 30 \right) \dots (ii)$$

$$(i)/(ii) \Rightarrow (t = 216 \text{ s})$$

21.



23. Let  $v_1$  and  $v_2$  be the velocities of the body just before and just after the collision.

$$KE_1 = \frac{1}{2} mv_1^2 = mgh_1 \dots (1)$$

$$KE_2 = \frac{1}{2} mv_2^2 = mgh_2 \dots (2)$$

$$\text{from (1) \& (2)} \Rightarrow \frac{v_1^2}{v_2^2} = \frac{h_1}{h_2} = \frac{10}{2.5} = 4$$

$$\Rightarrow \frac{v_1}{v_2} = 2$$

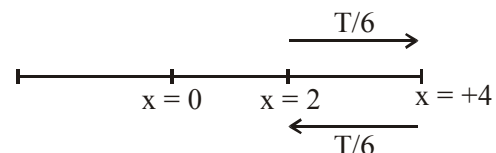
24. Heat gain = Heat loose

$$m_w s_w \Delta T = m_s L + m_w s_w (\Delta T)$$

$$40 \times 1 \times (54.3 - 25) = 2 \times L + 2 \times 1 \times (100 - 54.3)$$

$$L = 540.3 \text{ cal/g}$$

25.



$$\text{minimum time taken by particle} = \frac{T}{6} + \frac{T}{6}$$

$$= \frac{T}{3}$$

$$= 0.4 \text{ s}$$

26.  $H_{\max} = \frac{u^2 \sin^2 \theta}{2g}$

According to problem

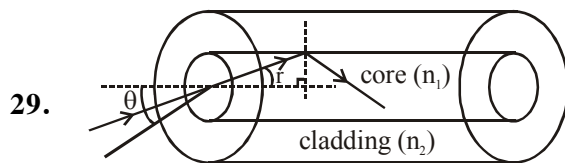
$$\frac{u_1^2 \sin^2 45^\circ}{2g} = \frac{u_2^2 \sin^2 60^\circ}{2g}$$

$$\Rightarrow \frac{u_1^2}{u_2^2} = \frac{\sin^2 60^\circ}{\sin^2 45^\circ} \Rightarrow \frac{u_1}{u_2} = \frac{\sqrt{3}/2}{1/\sqrt{2}} = \sqrt{\frac{3}{2}}$$

27. The sixth coin is under the weight of four coins above it. Hence,

Reaction of the 6<sup>th</sup> coin on the 7<sup>th</sup> coin

$$= \text{Force on the 6<sup>th</sup> coin due to 7<sup>th</sup> coin} = 4mg$$



$$1 \times \sin \theta = n_1 \times \sin r$$

$$\sin r = \frac{\sin \theta}{n_1} \quad \dots(i)$$

$$i > \theta_c$$

$$90 - r > \theta_c$$

$$\sin (90 - r) > \sin \theta_c$$

$$\cos r > \frac{n_2}{n_1} \quad \dots(ii)$$

on squaring and then adding equ. (i) and eq. (ii), we get

$$\frac{\sin^2 \theta}{n_1^2} + \frac{n_2^2}{n_1^2} \geq 1$$

$$\sin^2 \theta + n_2^2 \geq n_1^2$$

$$\sin^2 \theta \geq n_1^2 - n_2^2$$

$$\sin \theta \geq \sqrt{n_1^2 - n_2^2}$$

$$\theta \geq \sin^{-1} \sqrt{n_1^2 - n_2^2}$$

30.  $W = hv - \frac{1}{2}mv^2$

$hv$  = energy of incident photon

$$= \frac{12400}{1240} \text{ eV} = 10 \text{ eV}$$

$$\therefore W = 10 - 8 = 2 \text{ eV}$$

So,  $\lambda_0$  = Threshold wavelength

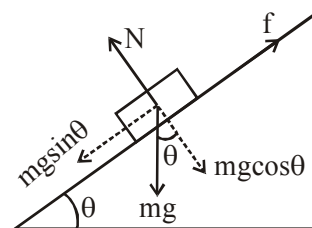
$$= \frac{12400}{2 \text{ eV}} \text{ \AA} = 6200 \text{ \AA}$$

31.  $R = \frac{u^2 \sin 2\theta}{g} = \frac{2u_x v_y}{g}$

$\therefore$  Range  $\propto$  horizontal initial velocity ( $u_x$ )

In path 4 range is maximum so football possess maximum horizontal velocity in this path.

32. The various forces acting on the block are as shown in the figure.



From figure

$$mg \sin \theta = f \quad \dots(i)$$

$$mg \cos \theta = N \quad \dots(ii)$$

Divide (i) by (ii), we get

$$\tan \theta = \frac{f}{N} = \frac{\mu N}{N}$$

$$\text{or } \theta = \tan^{-1}(\mu)$$

33.  $dq = I dt = I_0 (1 - e^{-t/T}) = \frac{E}{R} (1 - e^{-t/T})$

$$q = \int_0^T \frac{E}{R} (1 - e^{-t/T}) dt = \frac{E}{R} [t + T e^{-t/T}]_0^T$$

$$= \frac{E}{R} \left[ T + \frac{T}{e} - T \right] = \frac{E T}{R e}$$

34.  $f = 5 \text{ cm}, D = 25 \text{ cm}$

$$MP = 1 + \frac{D}{f} = 1 + \frac{25}{5} = 6$$

35. No of  $\alpha$  particles (n)  $\Rightarrow \frac{232-208}{4} = \frac{24}{4} = 6$

No of  $\beta$  particles :-  $z - 2n(\alpha) + n'(\beta) = 82$

$90 - 2(6) + n'(\beta) = 82$

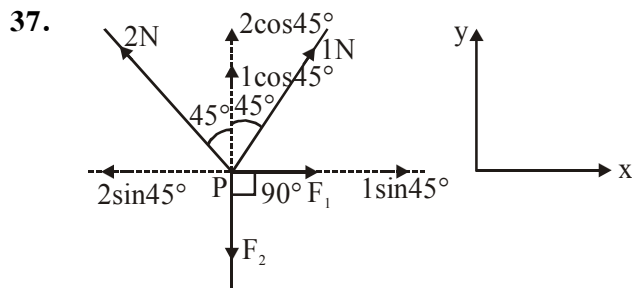
$n'(\beta) = 82 - 78 = 4$

36.  $P = \frac{V^2}{R}$  so,  $R = \frac{V^2}{P} \therefore R_1 = \frac{V^2}{100}$  &  $R_2 = R_3 = \frac{V^2}{60}$

Now,  $P_1 = \frac{(250)^2}{(R_1 + R_2)^2} \cdot R_1$  and

$P_2 = \frac{(250)^2}{(R_1 + R_2)^2} \cdot R_2$  and  $P_3 = \frac{(250)^2}{R_3}$

$P_1 : P_2 : P_3 = 15 : 25 : 64 \Rightarrow P_1 < P_2 < P_3$



Applying equilibrium conditions,

$\Sigma F_x = 0$

$\Rightarrow F_1 + 1\sin 45^\circ - 2\sin 45^\circ = 0$

or  $F_1 = 2\sin 45^\circ - 1\sin 45^\circ$

$= \frac{2}{\sqrt{2}} - \frac{1}{\sqrt{2}} = \frac{2-1}{\sqrt{2}} = \frac{1}{\sqrt{2}} \text{ N}$

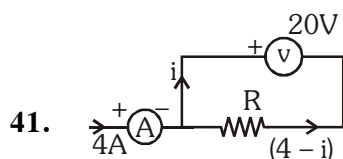
and  $\Sigma F_y = 0$

$\Rightarrow 1\cos 45^\circ + 2\sin 45^\circ - F_2 = 0$

$F_2 = \frac{2}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \frac{2+1}{\sqrt{2}} = \frac{3}{\sqrt{2}} \text{ N}$

38. B, v and I should be mutually perpendicular.

40. Velocity of a wave is independent of others.



$(4-i)R = 20$

$R = \frac{20}{(4-i)}$

for min. value of i ( $R > 5$ )

43. In Position-1

$\tan \frac{\pi}{6} = \frac{\omega L}{R} = \frac{(1000)(\sqrt{3} \times 10^{-3})}{R}$  or  $\frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{R}$

$\therefore R = 3\Omega$

In Position-2

$\tan \phi = \frac{X_C}{R} = \frac{(1/\omega C)}{R}$

$= \frac{1/\left(1000 \times \frac{1000}{3} \times 10^{-6}\right)}{3} = 1$

$\therefore \phi = \frac{\pi}{4}$

44. Velocity u of the body when it enters the liquid is given by

$mgh = \frac{1}{2}mu^2$  or  $u = \sqrt{2gh}$

Let

volume of the body = V

mass of the body = Vd

weight of the body = Vdg

Mass of liquid displaced = VD

Weight of liquid displaced = VDg

Net upward force = VDg - Vdg

$= Vg(D-d)$

Retardation =  $\frac{\text{net weight}}{\text{mass}}$

$= \frac{V(D-d)g}{Vd} = \left(\frac{D-d}{d}\right)g$

Acceleration  $a = -\left(\frac{D-d}{d}\right)g$

Final velocity, v in the liquid when the body is instantaneously at rest is zero. Let the time taken be t.

Now  $v = u + at$

$0 = \sqrt{2gh} - \left(\frac{D-d}{d}\right)gt, \left(\frac{D-d}{d}\right)gt = \sqrt{2gh}$

$\therefore t = \left[\frac{d}{D-d}\right]\sqrt{\frac{2h}{g}}$

45. Here, speed of sound,  $v = 340 \text{ ms}^{-1}$   
Length of pipe,  $L = 17 \text{ cm} = 17 \times 10^{-2} \text{ m}$   
In a closed pipe (open at one end), the frequency of its  $n^{\text{th}}$  harmonic is

$$v_n = \frac{nv}{4L} \quad \text{where } n = 1, 3, 5, 7, \dots$$

Let  $n^{\text{th}}$  harmonic of closed pipe resonates with 1.5 kHz source.

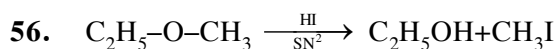
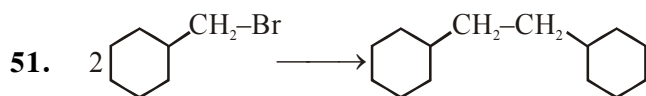
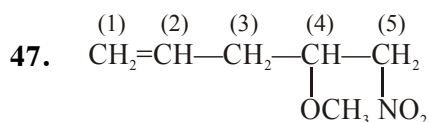
$$\therefore v_n = 1.5 \times 10^3 \text{ Hz}$$

$$\therefore 1.5 \times 10^3 = \frac{nv}{4L}$$

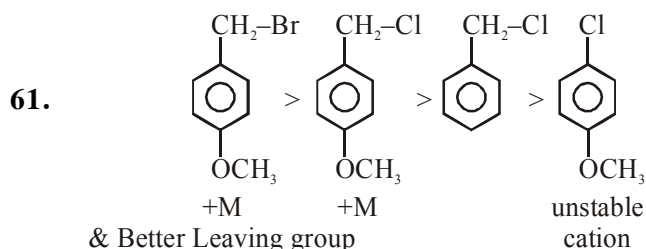
$$n = \frac{1.5 \times 10^3 \times 4 \times 17 \times 10^{-2}}{340}$$

$$n = 3$$

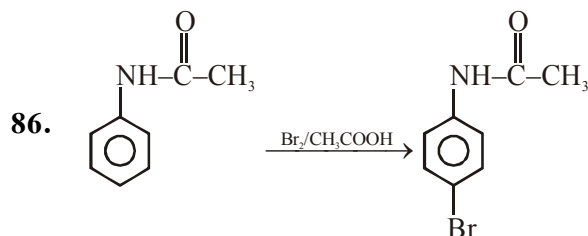
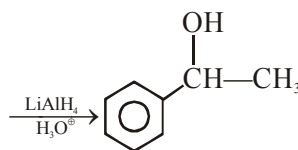
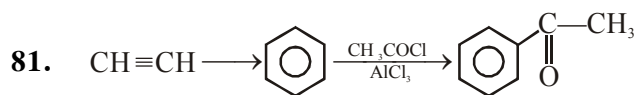
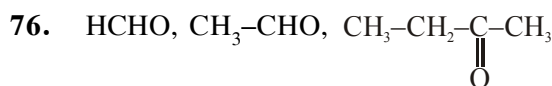
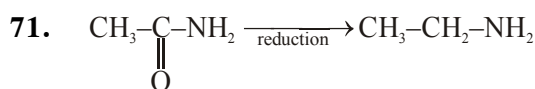
46. Birch reduction



57. Ref./NCERT/XII/Part-II/408



62. Ref./NCERT/XII/Part-II/413



92. NCERT XII<sup>th</sup> Pg.#60 Para-4.2

93. NCERT Pg.#38

94. Module Pg.# 51

95. NCERT XII Pg # 204-205, para 3

98. NCERT Pg # 259

102. NCERT Pg.#5

104. NCERT XII Pg # 72 para 3

107. Module # 3, Pg # 114

110. NCERT Pg # 307

112. NCERT Pg.#9

115. Module No.2, Page # 213 & 214

117. NCERT XII Pg # 133 (E)

122. NCERT Pg.#13

123. NCERT Pg.# 55, fig 4.16

124. NCERT XII Pg # 77, para 2

126. Either oligomycin or DCCD blocks the  $\text{H}^+$  leaf in membrane depleted to  $\text{F}_1$  thus oligomycin and DCCD inhibit the interacting with  $\text{F}_0$ .

(DCCD = N,N dicyclohexyl carbodiimide)

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| <p>127. Module # 3, Pg # 115</p> <p>130. NCERT Pg # 305</p> <p>132. NCERT Pg.#19</p> <p>133. NCERT Pg.# 113, 7.16</p> <p>134. NCERT XII Pg # 91 para 2</p> <p>135. Module No.2, Page # 35 &amp; 37</p> <p>136. In a fully turgid cell, difusion pressure dificit = zero as, turgor pressure = osmotic pressure</p> <p>141. NCERT XII<sup>th</sup> Pg.#47 Para-3.3</p> <p>142. NCERT Pg.#19</p> <p>143. NCERT Pg. # 113, 114</p> <p>144. NCERT XII Pg # 89, 90 para 2 fig 5.15</p> <p>146. NCERT Pg.# 233</p> <p>150. NCERT XI<sup>th</sup> Pg.#318 Para-1</p> <p>151. NCERT XII<sup>th</sup> Pg.#47 Para-3.6</p> | <p>156. (Chadwick and Burg proposed the effect of ethylene for positive geotrophic movement of plant)</p> <p>160. NCERT XI<sup>th</sup> Pg.#323,324,</p> <p>165. NCERT XI Pg # 165, para 1</p> <p>170. NCERT XI<sup>th</sup> Pg.#335,336</p> <p>171. NCERT XII<sup>th</sup> Pg.#51</p> <p>172. NCERT Pg.#35</p> <p>173. Module Pg.# 96</p> <p>174. NCERT XII Pg # 195, para 3</p> <p>175. NCERT XI Pg # 162, last para</p> <p>176. NCERT Pg.# 211</p> <p style="padding-left: 40px;">PS-I reaction center is P<sub>700</sub></p> <p>180. NCERT XI<sup>th</sup> Pg.#338 Para-2,3</p> |
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